Question 1. The following questions require true/false answers and justification. Full credit will only be given for correct answers with correct justification, but partial credit is possible.

(a) (true/false) Consider a federated averaging algorithm, where individual agents construct mini-batch gradient approximations. Suppose the step-sizes are normalized to ensure there is no bias, and the step-sizes are chosen small enough so that higher-order terms in the step-size can be neglected. Then, increasing the number of local updates by a factor of 10 will have approximately the same effect on the steady-state performance as increasing the mini-batch size by a factor of 10. [5 marks]

Solution. The statement is false. [2 marks]

The performance of the federated averaging algorithm is given by $\frac{\mu\sigma_{\text{fed}}^2}{\nu}$, where: [3 marks]

$$\sigma_{\text{fed}}^2 = \frac{1}{KL}\sum_{k=1}^{K}\left(2\frac{\beta_k^2}{E_k}\|w_k^o - w^o\|^2 + \frac{\sigma_k^2}{E_k}\right) + \frac{1}{KL}\frac{K-L}{K-1}\sum_{k=1}^{K}4\delta_k^2\|w_k^o - w^o\|^2$$

Observe here that the effect of the local updates and the mini-batch size are not the same, due to the presence of the heterogeneity term. It is only in the homogeneous case that we can simplify:

$$\sigma_{\text{fed}}^2 = \frac{1}{KL}\sum_{k=1}^{K}\frac{\sigma^2}{E} = \frac{1}{L}\frac{\sigma^2}{E}$$

(b) (true/false) The Metropolis-Rule is useful because it gives rise to a left-stochastic combination matrix. [5 marks]

Solution. The statement is false. [2 marks]

All combination policies are left-stochastic. The Metropolis-Rule gives rise to a symmetric, and hence doubly-stochastic, combination matrix. [3 marks]

(c) (true/false) When agents are homogeneous, the steady-state mean-squared deviations of the distributed gradient descent algorithm and the EXTRA algorithm are identical. [5 marks]

Solution. The statement is true. [2 marks]

The bias of penalty-based algorithms disappears when agents share a minimizer, which is the case when they are homogeneous. [3 marks]

(d) (true/false) The eigenvectors of the mixing matrix determine how quickly information diffuses over the graph. [5 marks]

Solution. The statement is false. [2 marks]

The eigenvalues determine the rate of convergence, the eigenvectors only determine the limiting point. [3 marks]

(e) (true/false) Consider the static consensus algorithm: [5 marks]

$$w_{k,i} = \sum_{\ell\in\mathcal{N}_k} a_{\ell k} w_{\ell,i-1}$$

Suppose all weights $w_{k,0}$ are initialized at the same point, i.e., $w_{k,0} = g$ for all k. Then, $\lim_{i\to\infty} w_{k,i}$ will be unaffected by A, as long as it is primitive and a valid combination matrix.

Solution. The statement is true. [2 marks]

The limiting point is given by $\sum_{k=1}^{K} p_k w_{k,0} = \sum_{k=1}^{K} p_k g = g$, which is independent of p and hence A. This is due to the fact that all agents were initialized at the same point. [3 marks]

(f) (true/false) We consider a collection of $K = 1000$ agents, and two architectures for distributed learning. In both cases, the agents are homogeneous, and the step-size is small enough so that higher-order terms in the step-size can be neglected. In architecture (i), the agents are running a federated averaging algorithm with the help of a fusion center. All agents perform $E = 10$ local updates, and $L = 10$ agents participate in each round. In architecture (ii), the agents operate in a decentralized manner and are running the EXTRA algorithm. Then, the two architectures will exhibit approximately the same steady-state performance. [5 marks]

Solution. The statement is false. [2 marks]

The performance of the federated averaging algorithm in the homogeneous setting is given by $O\left(\frac{\mu\sigma^2}{\nu LE}\right) = O\left(\frac{\mu\sigma^2}{100\nu}\right)$. The performance of the EXTRA algorithm is given by $O\left(\frac{\mu\sigma^2}{K\nu}\right) = O\left(\frac{\mu\sigma^2}{1000\nu}\right)$ and so the bounds differ by approximately a factor of 10. [3 marks]

(g) (true/false) Without proper step-size normalization by the number of local updates, the federated averaging algorithm will not converge, irrespective of how small the step-size was chosen. [5 marks]

Solution. The statement is false. [2 marks]

The federated averaging algorithm will converge even without step-size normalization, but the limiting point will be biased (in proportion to the number of local updates taken). [3 marks]

(h) (true/false) Consider a fully connected graph, where every agent is connected to every other agent (and itself). Then, the performance of the diffusion algorithm will coincide with the performance of the centralized stochastic gradient algorithm, as long as the combination matrix is chosen properly. [5 marks]

Solution. The statement is true. [2 marks]

The diffusion algorithm for a fully connected graph is equivalent to the centralized stochastic gradient algorithm, where every individual agent plays the role of a fusion center. [3 marks]

Question 2. We consider a collection of K agents, where each agent k is equipped with a local logistic regression risk:

$$J_k(w) \triangleq \mathbb{E} \ln\left(1 + e^{-\boldsymbol{\gamma}_k \boldsymbol{h}_k^{\mathsf{T}} w}\right) + \frac{\rho}{2}\|w\|^2$$

We are interested in solving the consensus optimization problem:

$$w^o = \arg\min_{w \in \mathbb{R}^M} \frac{1}{K}\sum_{k=1}^{K} J_k(w)$$

(a) Formulate the recursions for the federated averaging algorithm, when agents are performing local mini-batch gradient approximations. Suppose that at time i and local step e, each agent k observes B_k pairs $\{\boldsymbol{h}_{k,i,b,e}, \boldsymbol{\gamma}_{k,i,b,e}\}$. All observations (across space and time) are independent. [5 marks]

Solution. We have for $e = 1, \ldots, E_k$: [5 marks]

$$\boldsymbol{\phi}_{k,e} = \left(1 - \frac{\mu\rho}{E_k}\right)\boldsymbol{\phi}_{k,e-1} + \frac{\mu}{E_k B_k}\sum_{b=1}^{B_k} \frac{\boldsymbol{\gamma}_{k,i,e,b}\boldsymbol{h}_{k,i,e,b}}{1 + e^{\boldsymbol{\gamma}_{k,i,e,b}\boldsymbol{h}_{k,i,e,b}^{\mathsf{T}}\boldsymbol{w}_{k,e-1}}}$$

Then:

$$\begin{aligned}\boldsymbol{\psi}_{k,i} &= \boldsymbol{\phi}_{k,E_k} \\ \boldsymbol{w}_i &= \frac{1}{L}\sum_{k\in\mathcal{L}_i} \boldsymbol{\psi}_{k,i}\end{aligned}$$

(b) Determine an expression for the steady-state mean-squared deviation of the algorithm developed in (a). [10 marks]

Solution. We need to find: [4 marks]

$$\sigma_{\text{fed}}^2 = \frac{1}{KL}\sum_{k=1}^{K}\left(2\frac{\beta_k^2}{E_k}\|w_k^o - w^o\|^2 + \frac{\sigma_k^2}{E_k}\right) + \frac{1}{KL}\frac{K-L}{K-1}\sum_{k=1}^{K} 4\delta_k^2\|w_k^o - w^o\|^2$$

For logistic regression with mini-batch gradient approximations: [4 marks]

$$\begin{aligned}\beta_k^2 &= 0 \\ \sigma_k^2 &= \frac{\text{Tr}(R_{\boldsymbol{h}_k})}{B_k} \\ \delta_k^2 &= \rho + \lambda_{\max}(R_{\boldsymbol{h}_k}) \\ \nu &= \rho\end{aligned}$$

Plugging everything into $\limsup_{i\to\infty}\mathbb{E}\|\widetilde{\boldsymbol{w}}_i\|^2 = O\left(\frac{\mu\sigma_{\text{fed}}^2}{\nu}\right)$ yields the result. [2 marks]

(c) Formulate the recursions for the AugDGM algorithm, when agents are performing local mini-batch gradient approximations. [5 marks]

Solution. We have: [5 marks]

$$\boldsymbol{\psi}_{k,i} = (1 - \mu\rho)\boldsymbol{w}_{k,i-1} + \mu\frac{\boldsymbol{\gamma}_{k,i}\boldsymbol{h}_{k,i}}{1 + e^{\boldsymbol{\gamma}_{k,i}\boldsymbol{h}_{k,i}^{\mathsf{T}}\boldsymbol{w}_{k,i-1}}}$$

$$\boldsymbol{w}_{k,i} = \sum_{\ell \in \mathcal{N}_k} a_{\ell k} \boldsymbol{\psi}_{\ell,i}$$

(d) Determine an expression for the steady-state mean-squared deviation of the algorithm developed in (c). [10 marks]

Solution. The performance bound for the AugDGM algorithm is: [5 marks]

$$\text{MSD} = O\left(\frac{\mu\sigma^2}{\nu}\right) + O\left(\frac{\mu^2 K \lambda_2^2 \sigma^2}{1-\lambda_2}\right)$$

Having found the relevant constants in part (b), we can also plug them into this relation and obtain the result. [5 marks]

Question 3. We consider a collection of K, agents, where each agent k is equipped with a local least-squares risk:

$$J_k(w) \triangleq \frac{1}{2}\mathbb{E}\left\|\boldsymbol{\gamma}_k - \boldsymbol{h}_k^{\mathsf{T}} w\right\|^2$$

The local data follows the linear model:

$$\boldsymbol{\gamma}_k = \boldsymbol{h}_k^{\mathsf{T}} w^o + \boldsymbol{v}_k \tag{1}$$

The features $\boldsymbol{h}_k$ and $\boldsymbol{v}_k$ are normally distributed according to $\boldsymbol{h}_k \sim \mathcal{N}(0, R_{\boldsymbol{h}_k})$ and $\boldsymbol{v}_k \sim \mathcal{N}(0, \sigma_{\boldsymbol{v}_k}^2)$ respectively. The data is independently sampled across agents. Note that we are assuming that agents observe data generated by the same set of parameters w^o, albeit with features and noise with potentially differing statistics. It can be verified that the common model w^o in (1) is the minimizer of the local objectives for all k:

$$w^o = \arg\min_{w \in \mathbb{R}^M} J_k(w)$$

The agents run a variant of the diffusion algorithm, where there are C rounds of communication in between every gradient step. Here, $C \geq 1$ is a natural number. Specifically, each agent maintains a weight vector $\boldsymbol{w}_{k,i,c}$, where the indices k and i correspond to the agent and iteration index respectively, and c indexes the communication round. At time i, each agents first performs:

$$\boldsymbol{\psi}_{k,i} = \boldsymbol{w}_{k,i-1,C} + \mu \boldsymbol{h}_{k,i}\left(\boldsymbol{\gamma}_{k,i} - \boldsymbol{h}_{k,i}^{\mathsf{T}} \boldsymbol{w}_{k,i-1,C}\right) \tag{2}$$

Then, each agent k initializes $\boldsymbol{w}_{k,i,0} = \boldsymbol{\psi}_{k,i}$ and performs C rounds of consensus averaging for $c = 1, \ldots, C$:

$$\boldsymbol{w}_{k,i,c} = \sum_{\ell \in \mathcal{N}_k} a_{\ell k} \boldsymbol{w}_{\ell,i,c-1} \tag{3}$$

After completing C exchanges, agent k now has $\boldsymbol{w}_{k,i,C}$, and the process repeats. Observe that by setting $C = 1$, we recover the classical diffusion algorithm we studied in the lectures. The graph is strongly-connected and the combination matrix A is symmetric, so that $A = A^{\mathsf{T}}$ is primitive.

(a) Argue that (2)–(3) is equivalent to the classical diffusion algorithm after replacing the combination matrix A by a suitably chosen A', i.e.: [5 marks]

$$\begin{aligned}
\boldsymbol{\psi}_{k,i} &= \boldsymbol{w}_{k,i-1,C} + \mu \boldsymbol{h}_{k,i}\left(\boldsymbol{\gamma}_{k,i} - \boldsymbol{h}_{k,i}^{\mathsf{T}} \boldsymbol{w}_{k,i-1,C}\right) \\
\boldsymbol{w}_{k,i,C} &= \sum_{\ell \in \mathcal{N}_k} a'_{\ell k} \boldsymbol{\psi}_{\ell,i}
\end{aligned}$$

Determine an expression for A' in terms of A.

Solution. Taking C consensus steps is equivalent to taking the C-th power of the combination matrix. Hence, $A' = A^C$. [5 marks]

(b) Show that A' is primitive, symmetric and doubly-stochastic. Find an expression for its mixing rate in terms of the mixing rate of A. [10 marks]

Solution. If $A = U\Lambda U^{\mathsf{T}}$, we know that $A' = A^C = U\Lambda^C U^{\mathsf{T}}$. Hence, $\lambda_2(A') = \lambda_2(A)^C$ [5 marks]

Symmetry is immediate from symmetry of A, and to verify right-stochasticity, note that $A'\mathbb{1} = A^C\mathbb{1} = \mathbb{1}$. [5 marks]

(c) Determine a bound for the steady-state performance of the network, i.e.: [10 marks]

$$\limsup_{i\to\infty} \frac{1}{K}\sum_{k=1}^{K} \mathbb{E}\|\widetilde{\boldsymbol{w}}_{k,i,C}\|^2$$

Solution. The performance of the diffusion algorithm is given by: [4 marks]

$$\text{MSD} = O\left(\frac{\mu\sigma^2}{\nu}\right) + O\left(\frac{\mu^2 b}{K(1-\lambda_2)}\right)$$

We have: [4 marks]

$$\begin{aligned}
\nu &= \lambda_{\min}\left(\sum_{k=1}^{K} p_k R_{\boldsymbol{h}_k}\right) \\
\sigma^2 &= \frac{\sum_{k=1}^{K}\sigma_{\boldsymbol{v}_k}^2 \operatorname{Tr}\left(R_{\boldsymbol{h}_k}\right)}{K^2} \\
b &= O(\lambda_2^2 K^2 \sigma^2)
\end{aligned}$$

The simplified result for b follows from the fact that the models are common. Plugging everything together gives the result. [2 marks]

(d) Determine a bound on C, such that the difference between the performance derived in part c) and the performance of the centralized stochastic gradient algorithm (i.e., the cost of decentralization) is less than ϵ. [5 marks]

Solution. We need [5 marks]

$$O\left(\frac{\mu^2\lambda_2(A)^{2C}K^2\sigma^2}{K(1-\lambda_2(A))}\right) \le O\left(\mu^2\lambda_2(A)^{2C}K\sigma^2\right) \le \epsilon \iff \lambda_2(A)^{2C} \le O\left(\frac{\epsilon}{\mu^2 K\sigma^2}\right)$$